$$[T]_{SS'} = \begin{pmatrix} -2 & -6 & 2 \\ 2 & 5 & -2 \\ 2 & 4 & -1 \end{pmatrix}$$

Theorem:-

Let T be a LT on a finite dimensional v.s V and let S be a basis of V, then for any vector $v \in V$,

$$[T]_S [v]_S = [T(v)]_S$$

Change of basis matrix from S to S' :-

Let $S = \{u_1, u_2, \ldots, u_n\}$

$S' = \{v_1, v_2, \ldots, v_n\}$

$v_1 = a_{11}u_1 + a_{12}u_2 + \ldots + a_{1n}u_n$

$v_2 = a_{21}u_1 + a_{22}u_2 + \ldots + a_{2n}u_n$

$\vdots$

$v_n = a_{n1}u_1 + a_{n2}u_2 + \ldots + a_{nn}u_n$

$[v_1]_S = (a_{11}, a_{12}, \ldots, a_{1n})^T$

$\vdots$

$[v_n]_S = (a_{n1}, a_{n2}, \ldots, a_{nn})^T$

$$P = \begin{bmatrix} a_{11} & a_{21} & \ldots & a_{n1} \\ a_{12} & a_{22} & \ldots & a_{n2} \\ \vdots & & & \\ a_{1n} & a_{2n} & \ldots & a_{nn} \end{bmatrix}$$

← Transition matrix

Change of basis matrix from S' to S is $Q = P^{-1}$.

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